

Building a Job Ads Database from Swedish Digitized Newspapers

Historical Labor Markets in Text | University of Graz | 25 April 2026

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This talk is based on *The Job Ads Database (JADB) Prototype* (Lund Papers in Economic History No. 264, 2026) — co-authored with Peter Gladoić Håkansson, Tobias Karlsson and Matti la Mela

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Today I will cover:

1. The problem motivating JADB
2. Data, method, and preliminary findings
3. Obstacles and limitations
4. Epistemology

The problem

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- Women's work, domestic service, youth employment
- Census records, tax rolls, and guild registers show us the *formal* labour market
- Digitized newspapers are a *promising* complement to the formal labour market in Sweden.

What is JADB?

A database of job advertisements extracted from digitized Swedish newspapers

The prototype (this paper)

- *Dagens Nyheter*, 1870–1970
- One newspaper, consistent coverage
- ~2 million page hits
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The ambition

- All digitized Swedish newspapers
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Search vocabulary: 150,554 occupation title spellings from Swedish HISCO

HISCO as search vocabulary

The Historical International Standard Classification of Occupations

- Swedish version assembled by SwedPop: 179,326 spelling variants
- After filtering: **150,554 actual occupation titles**
- Each title classified by gender (male / female / both)

Gender	Spellings	Occupations
Man	71,123	43,343
Woman	33,469	18,851
Both	45,962	6,346
Total	150,554	68,530

Recalibrated distribution. Source: Edvinsson & Westberg (2024); SwedPop (2022)

The corpus

KB's digitized newspaper database: ~2.5 million issues

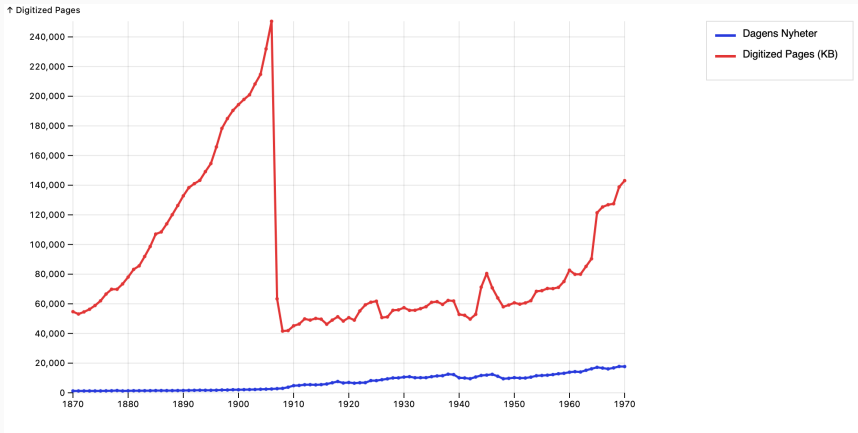


Figure 1: Coverage of Dagens Nyheter vs. all KB newspapers, 1870–1970. Source: jobadsdata.github.io/jadb-1-timelines

The proximity search

No article boundaries → we search for *proximity*

"<OCCUPATION> <MARKER>"~5

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→ finds pages where *springflicka* appears within 5 words of *SÖKES*, *PLATS*, *ERHÅLLES*

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Example: "springflicka SÖKES"~5 OR "springflicka PLATS"~5 OR "springflicka ERHÅLLES"~5

→ finds pages where *springflicka* appears within 5 words of *SÖKES*, *PLATS*, *ERHÅLLES*

What we measure: number of pages with at least one match per year

Data quality and access constraints

This approach has known noise sources:

- **OCR quality** varies by period, typeface, and print condition

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Manual checks suggest high precision — but these constraints make **transparency and reproducibility** essential

We built a reusable tool: **kb-trend**

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Anyone can replicate our searches or run their own

pypi.org/project/kb-trend | github.com/DigitalHistory-Lund/kb-trend

Limitations

What we can and cannot measure

We can measure

- Long-term trends in job ad frequency
- Gender distributions (via HISCO)
- Relative changes over time
- Signals of labour market activity

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- Number of individual ads
- Number of actual vacancies
- Ad content (wages, requirements)
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High-volume occupations are **undercounted**: many ads per page = one page hit

These timelines are **proxies** — signals, not direct counts

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The cost of advertising

Announcement prices in *Dagens Nyheter* (3-line ad, lowest category)

Year	Price (SEK)	Real price (1929 SEK)
1890	0.36	0.74
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Who could afford to advertise? (Bi et al. 2025: higher prices reduced ads by job seekers, especially women)

Gendered job ad timelines

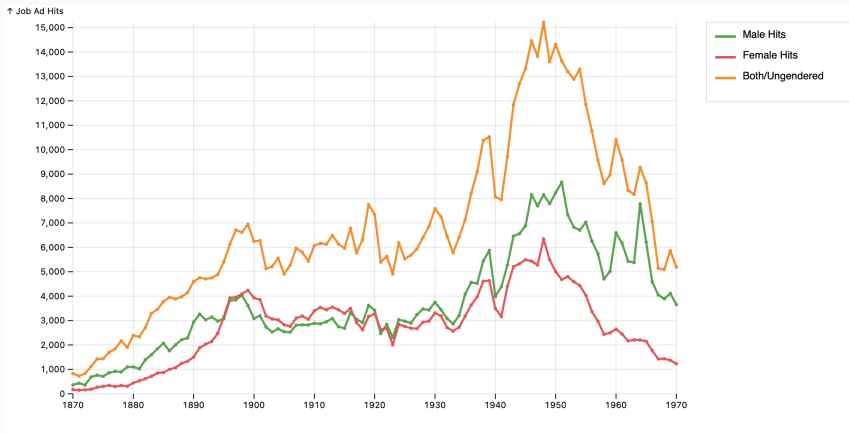


Figure 2: Gendered job title hits in Dagens Nyheter, 1870–1970. Source: jobadsdata.github.io/jadb-1-timelines

What explains the trends?

Job ad frequency is shaped by more than labour demand

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The newspaper as a recruitment channel had a **life cycle**

Comparison with SPEA data

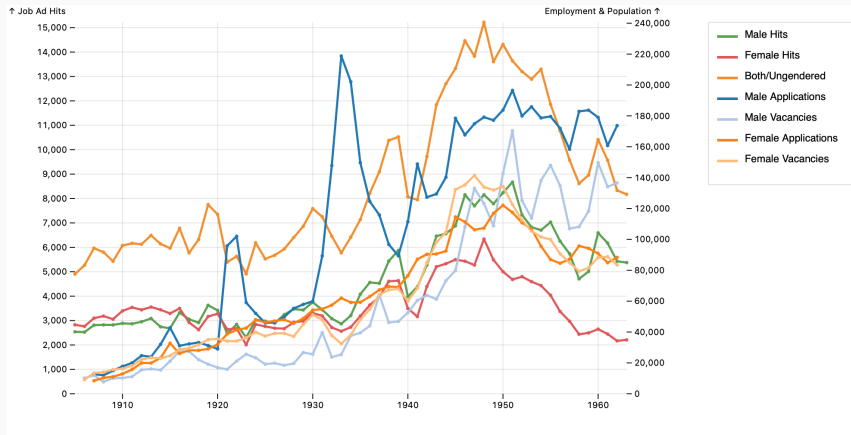


Figure 3: JADB data and Stockholm Employment Agency statistics, 1905–1963. Source: jobadsdata.github.io/jadb-1-timelines

The newspaper as a gendered space

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Different channels served different segments of the labour market

The “both” category

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Some of the largest “both”-coded occupations:

- **Hushållerska** (housekeeper) — linguistically feminine (-ska), coded as gender-neutral
- **Kontorist** (office clerk) — genuinely used for both sexes
- **Ingenjör** (engineer) — grammatically unmarked
- **Lärarinna** (teacher) — linguistically feminine (-inna), coded as gender-neutral

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The “both” category is not simply “ungendered” — it contains occupations with **feminine linguistic forms** classified as gender-neutral by HISCO

This complicates the gender disaggregation

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Employer seeking worker

Short text, occupation title, sometimes address or telephone number, markers like *sökes* or *plats*

Worker seeking position

Often even shorter, sometimes just a title and *söker plats* (seeks position)

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Our hermeneutic readings support the hypothesis that proximity-search hits are genuine job ads, and most are employers seeking workers

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The KB API returns annual counts — but the underlying data has **dates**

A future version could map seasonal patterns across occupations and decades

Doing research at the mercy of infrastructure

A broader challenge for computational history

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A broader challenge for computational history

- Digitized corpora are **research infrastructure** — but not always designed for research use
- APIs may be undocumented, unversioned, rate-limited, or change without notice
- Coverage decisions (what gets digitized, when) are made for institutional reasons, not research ones

Expanding JADB in **width** and **depth**

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- **Width:** successively adding more newspapers from the KB corpus, developing normalization methods for inconsistent coverage
- **Depth:** disaggregating page hits into individual job advertisements — layout analysis, ad boundary detection, text extraction
- **Analysis:** mapping occupational titles, gender distributions, and job descriptions across all digitized Swedish newspapers from the past 200 years

How we work: transparency by design

Division of labour

Data scraping, cleaning, aggregation → static websites on GitHub Pages

Co-authors explore and analyse through the public dashboards — on their own terms

Why public and static?

- Half-open sites are expensive

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The research tool **is** the public output

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Interactive dashboards

- jobadsdata.github.io/jadb-1-timelines — full timeline explorer
- jobadsdata.github.io/HISCOTrend — browse by HISCO code

Tool

- pypi.org/project/kb-trend — run your own searches

Paper

- Lund Papers in Economic History No. 264 (2026)

This is designed to be **used by others**

Explore the data, export it, build on it

1. Digitized newspapers are a promising but underexplored source for the Swedish historical labour markets — even coarse page-hit data yields usable signals
2. Even coarse-grained page-hit data reveals meaningful long-term trends in job advertising, by occupation and gender
3. The most interesting findings are often in the *divergences* — between sources, between genders, between channels

Thank you

Mathias Johansson — Lund University

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Matti la Mela — Uppsala University

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